One Number Can't Tell 8 Billion Stories

Births, deaths and migration in 237 countries and areas — UN estimates 1950–2023, projections to 2100 — and where the UN should look next

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METHODS AT A GLANCE **survival analysis · curve registration · fixed-effects panel regression · Moran's I · principal component analysis · k-nearest neighbours · Monte Carlo · k-means clustering · block bootstrap**

130 of 237 below replacement fertility (≈ 2.1 children per woman) · 84 still growing (66 births > deaths · 18 only because more arrive than leave)

Q1 How do births, deaths and migration relate — and how does that differ by region, country and income? P7 · P4 · **Q2** Does one indicator tell the whole story — or only several together? P8 · P5 · **Q3** Do countries share one road at different speeds — and what should the UN watch? P6 · P1 · P1–P8 = the eight suggested starting questions of the DataQuest Getting Started guide (§6), in order; P2 partly, P3 not addressed (both optional)

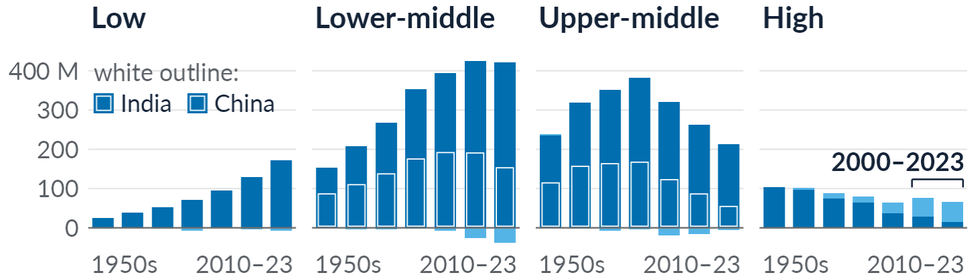
Which lever moves each country's population: births, deaths or migration?

Population change = births − deaths + net migration

Lever = a component of the population balance, not a policy switch.

High-income growth since 2000: mostly net migration

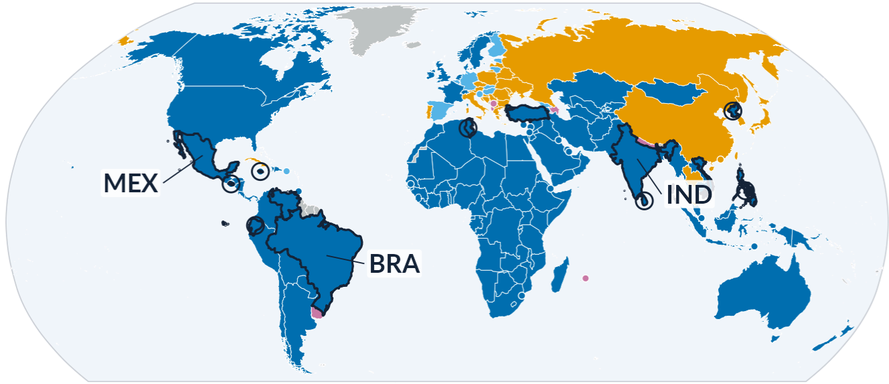
millions of people per decade · births − deaths · net migration



Sums of countries · World Bank FY2027 groups, all years · 2010–23 scaled to 10 years

Today's high-income countries (World Bank FY2027), 2000–2023: net migration ≈ 71 % of population change

64–78 % with other income lists and periods · a group total: it does not mean migration offsets low births country by country



Under 1 M people · 76 countries & areas

Latin Am. & Caribbean · 27

Oceania · 20

Europe & N. America · 17

Africa · 8

Asia · 4

2023 estimates · each country or area counted once

Growing: births > deaths	**165**	Below replacement, growing from births with no net arrivals: **22** countries, **2.2** bn people — largest: IND BRA MEX
Growing only because more arrive than leave	**18**	DEU ESP BEL
Shrinking: deaths > births	**35**	CHN RUS JPN
Shrinking: more leave than arrive	**19**	NPL AZE URY

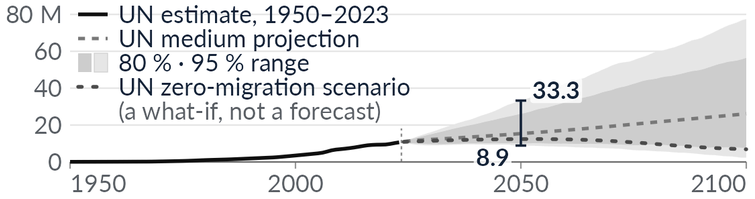
Light blue: 18 in 2023 (15–25 depending on the year) · grey: small states (< 1 M) · Map: Natural Earth; boundaries and names do not imply UN endorsement · examples chosen by rule: the three largest countries of each type (≥ 1 M); for blue, of these 22

UAE 2023: net migration +28.2 per 1,000 people · growth 3.70 % a year · fertility 1.20 children per woman

UAE = the twin picked by rule in Question 2 · net migration is often inferred from census gaps

UN medium projection, 2050: 15.4 M · 95 % range 8.9–33.3 M

95 % range = likely values from the UN's **probabilistic projections** · for 77 of 161 countries it spans both growth and decline by 2050

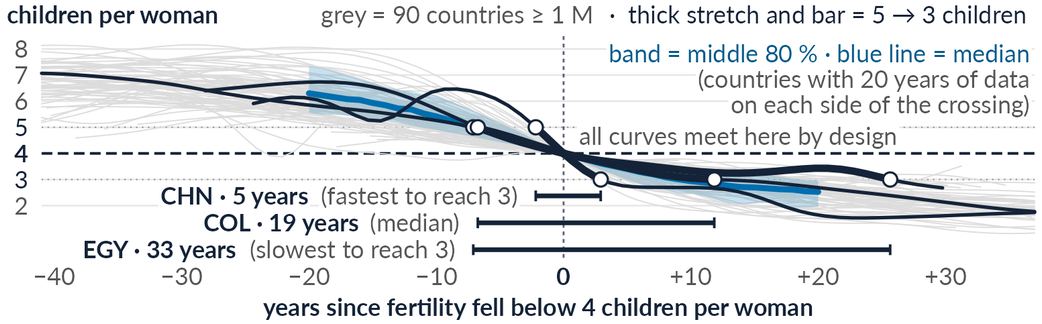


Today's high-income countries: about seven in ten of growth since 2000 came from net migration; elsewhere, from births. In 18 countries and areas, deaths already outnumber births and net arrivals keep them growing.

QUESTION 1 · PART 1

Same road, different speed: child mortality fell first, fertility followed

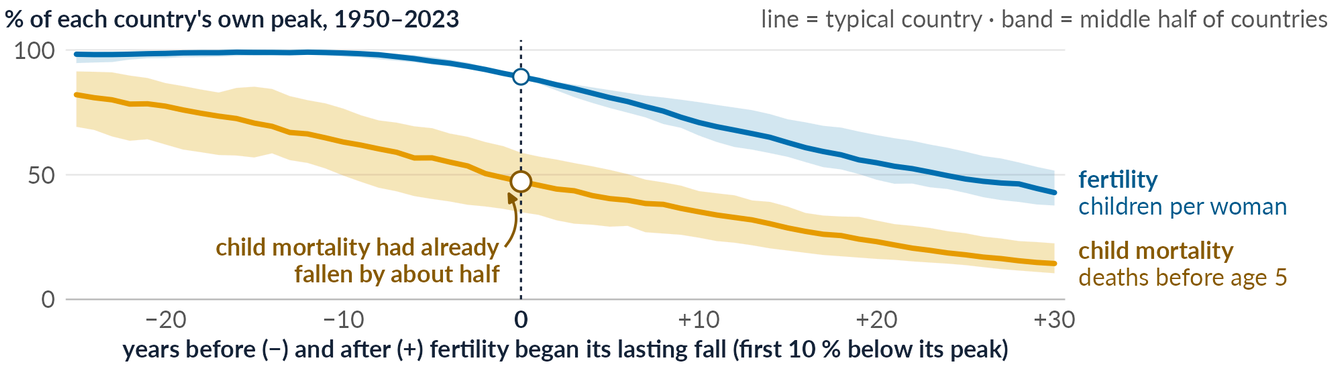
METHOD **Curve registration** (functional data) **aligning each country on the year its fertility fell below 4 removes 80 % of the variation between them**



5 → 3 children per woman: China 5 years · Egypt 33 years (fastest and slowest of countries ≥ 10 M that made the trip)

26 countries, mostly in sub-Saharan Africa, had not fallen below 4 by 2023 and are not shown; 45 were already below 4 in 1950

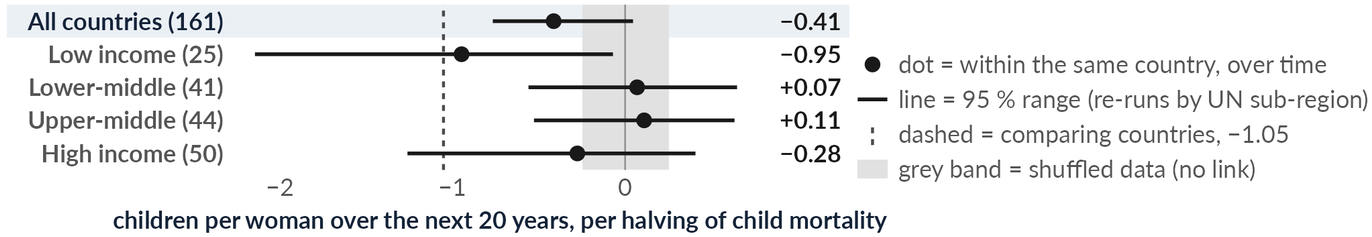
METHOD **Kaplan–Meier survival analysis** **child mortality began a lasting fall first in 106 of 109 high-fertility countries (97 %; 5+ children in 1950), typically at least 22 years earlier**



Fertility began to fall: ≈ 1970 in today's upper-middle-income countries · 1984 lower-middle · 2000 low-income

when it did, child mortality had already fallen by about half (50 %) · it was already falling in 1950 in 92 of the 109, so the true gap is probably longer · income groups = World Bank FY2027 (today's groups applied to all years)

METHOD **Fixed-effects panel regression** **within the same country the link is smaller: halving child mortality goes with −0.41 children per woman over 20 years, vs −1.05 comparing countries**



clearest in sub-Saharan Africa (−1.20); in Europe & N. America it goes the other way (+0.39) · association, not cause

Migration: a short memory and almost no neighbourhood

METHOD **Correlation over decades · Moran's I** — 0 = none, 1 = identical

same country, decade to decade: births & deaths ≥ **0.948** · migration **0.32**–**0.69** (latest 0.69)

neighbouring countries: fertility **0.74** · migration **0.11** (−0.03 without the Gulf)

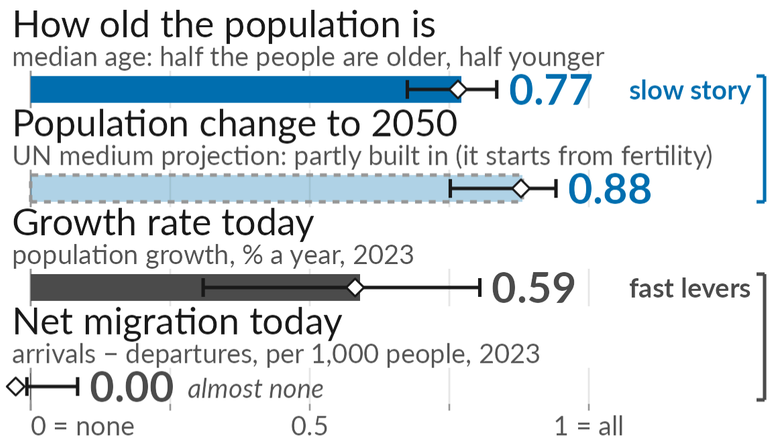
Net migration = arrivals − departures, often inferred from census gaps

QUESTION 1 ANSWERED
P7 · P4

Child deaths fell first, fertility followed — at each country's pace, latest in low-income countries; the link is clearest in sub-Saharan Africa. Migration moves on its own and accounts for ≈ seven in ten of high-income growth since 2000.

Does one indicator tell the whole story — or only several together?

METHOD **Adjusted R^2** · cross-validated **Fertility alone tells how old a population is, part of today's growth, almost nothing of migration**



R^2 · thin line = plausible range (re-runs by UN subregion)
◇ = same model on countries left out · exploratory, not causal

adjusted R^2, 161 countries ≥ 1 M, 2023 · R^2 = share of the differences between countries that fertility alone reproduces (0 = none, 1 = all) · 2050 change is partly built in

METHOD **k-nearest neighbours** · 5 look-alikes **Look-alikes on 12 indicators: fertility alone finds 13 % of them · the best single summary (axis 1): 20 % · both axes: 43 % · picking at random: 3 %**

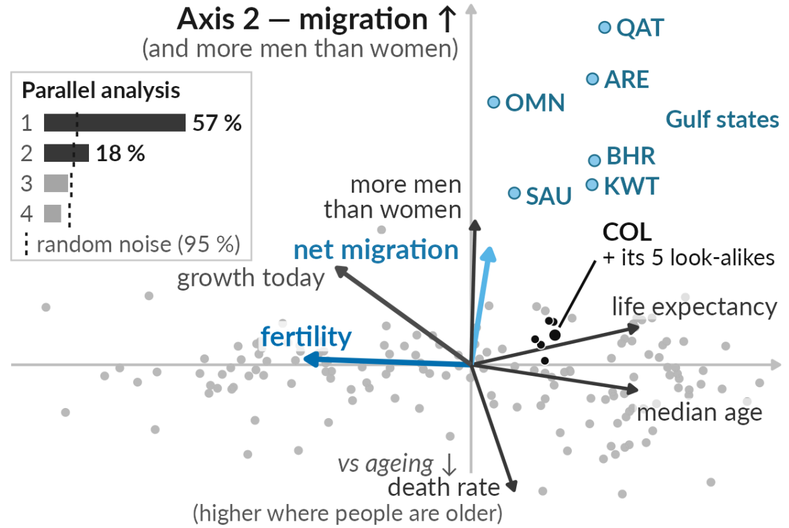
LOOK-ALIKES · COLOMBIA

by fertility it looks like
FRA AUS TUR USA NZL
by 12 indicators, like
IRN PER ECU PAN TUN
Colombia (our home country) as an illustration; all countries in the repository

SAME FERTILITY, OPPOSITE GROWTH · BY RULE

UAE · Belarus
same fertility **1.20** vs **1.21** children per woman → growth
▲ **+3.70 %** vs ▼ **−0.65 %** a year
twins chosen by rule from 12,880 pairs (≥ 1 M): same fertility (± 0.05), most different growth

METHOD **Principal component analysis** **only 2 axes stand out from random noise · fertility ≈ axis 1 (r = −0.94) · blind to axis 2 (r = 0.03)**



161 countries ≥ 1 M, 2023 · 12 indicators · exploratory, not causal
migration capped at ±3 SD for a few countries (incl. UAE, Oman)
arrows: 7 of the 12 indicators · axes not to the same scale

axis = one summary of all 12 indicators · r = correlation, −1 to 1 · axis 1 = 57 % of the variation (46–77 % with other indicator sets) · axis 2 = 18 %

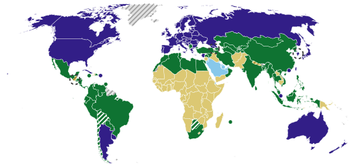
COMPOSITE INDEX **METHOD** **Monte Carlo** — 2,000 reasonable ways to build it = one score blending 5 indicators: weights, not formulas, decide — **one country in five can move more than 20 places** · a transition-stage index, not a ranking of development or well-being

QUESTION 2 ANSWERED
P8 · P5

No. Fertility alone tells age (0.77) and part of growth (0.59), not migration. Read together, transition and migration find 43 % of look-alikes; fertility alone, 13 %. In one blended score, weights move one country in five 20+ places.

One road, four groups — and what to watch on each

k-means (k = 4) on each country's 1950–2023 path in fertility, life expectancy, median age and net migration (decade averages) · re-checked with 500 random re-runs and neighbours



▨ changes group in > 1 in 5 re-runs (ringed if small) ▨ < 1 M people

Four groups of 70-year paths (k-means), countries ≥ 1 M
group · countries · people · three largest (by rule)

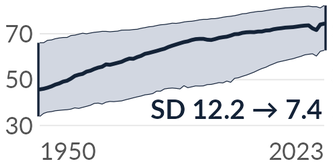
Early movers	50 · 1.4 bn	USA	RUS	JPN
Fast movers	51 · 4.9 bn	IND	CHN	IDN
Recent movers	53 · 1.7 bn	PAK	NGA	ETH
Migration-led	7 · 0.1 bn	SAU	JOR	ARE

Two in three pairs of neighbouring countries share a group (by chance: one in three) · Four in ten countries sit between two groups — a spectrum, not boxes

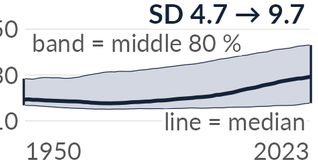
96 % of countries keep their group in at least 80 % of 500 re-runs · median age 30 → 40: Fast movers 32 years · Early movers 37 years (grey = UN medium projection after 2023, which assumes the transition continues)

Closer in life expectancy (SD 12.2 → 7.4 years), further apart in age (SD 4.7 → 9.7) and, in relative terms only, in child deaths: highest tenth 6× the lowest in 1950, 20× in 2023 (peak ≈ 27×, 2000s); the absolute gap shrank

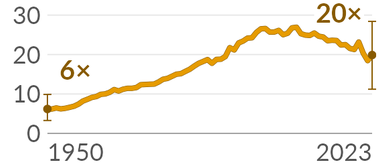
Life expectancy: closer



Median age: further apart



Child deaths: highest ÷ lowest tenth



SD = typical gap between countries (161 countries ≥ 1 M) · bars = 95 % ranges · but at 65 countries drifted apart: years left to live at 65, SD 1.47 → 2.85 · estimates 1950–2023

COUNTRIES WHERE…	ROAD (of the row's countries ≥ 1 M)	WATCH / PREPARE	FOR (MANDATE)	WHEN
child deaths off pace for SDG 3.2: 58 of 70 above 25 per 1,000 projected to stay above it in 2030 (UN medium); 8 on pace	mostly Recent movers (50 of 62)	child survival, newborn care; voluntary family planning; births to adolescent girls	UNICEF · WHO · UNFPA (SDG 3.2, 3.7 · ICPD)	by 2030
below replacement, growing from births with no net arrivals: 22 countries, 2.2 bn	all Fast movers (13 of 13)	34–38 more years of projected growth (UN medium) for most of their 2.2 bn people: pensions and long-term care now; no alarm about birth rates	UN DESA · WHO · ILO (Decade of Healthy Ageing · SDG 1.3)	2030–2060
deaths outnumber births: 53 countries (28 % of humanity); 106 projected by 2050 (UN medium)	mostly Early movers (33 of 35)	healthy longevity; measure mortality with life expectancy, not the crude death rate	WHO · UN DESA (SDG 3.4)	now–2040
growth comes only from arrivals: 18 countries and areas	all Early movers (11 of 11)	migrant inclusion; better arrival and departure data; read UN migration projections as scenarios	IOM · UN DESA (GCM obj. 1 & 16 · SDG 10.7)	now
shrinking as more leave than arrive (births > deaths): Nepal, Azerbaijan and Uruguay … · and everywhere: censuses and civil registration	mostly Fast movers (4 of 6)	decent work, safe return, diaspora links; publish lever types with their sensitivity ranges	IOM · ILO · UNSD (GCM obj. 2 & 19 · SDG 8.8, 17.19)	now / 2030 census round

How we kept it honest · ASA Ethical Guidelines (2022) We describe patterns; we do not claim causes.

Scenarios, not forecasts B.2 · C.4 **Weakest data, said aloud** B.1 · D.10 **Patterns, not causes** C.2 · C.3 **Levers, not verdicts** A.8 · D.6 **Open & checkable** B.3 · F.4

QUESTION 3 ANSWERED P6 · P1 **Mostly one road at different speeds: three stretches plus a migration-led group. Closer in life expectancy, further apart in age — and each group has something different to watch.**

References

DATA

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- UN DESA (2024). World Population Prospects 2024: Methodology of the United Nations population estimates and projections. UN DESA/POP/2024/DC/NO.10 · Summary of Results. UN DESA/POP/2024/TR/NO.9 · Release note on 2022 vs 2024 revisions (11 Jul 2024) · WPP 2024 Interim Update (Togo), 19 Jan 2026.
- WPP 2024 Supplement (births by sex, LE60) — prepared by the DataQuest organisers from the UN Data Portal.
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- Natural Earth (public domain) map data. Boundaries and names shown do not imply official endorsement or acceptance by the United Nations.

SOFTWARE

- R 4.6.1 (R Core Team, 2026); tidyverse (dplyr, tidyr, readr, ggplot2), sf, rnaturalearth, arrow, fixest, survival, fda, cluster, clue, mclust, boot, tweenr, ragg, ggflags, ggrepel, patchwork · FFmpeg (H.264) · reveal.js 6 (MIT) · flag-icons (MIT) · Lato (SIL OFL) · Playwright + Chromium (recording)

ACRONYMS

- SDG = UN Sustainable Development Goal target · ICPD = Cairo Programme of Action (1994) · GCM = Global Compact for Migration · UNSD = UN Statistics Division.

ACKNOWLEDGEMENTS

- Questions P1–P8: ASA DataQuest 2026 Getting Started guide (UN WPP 2024), section 6 "Suggested starting questions", numbered in order. Before designing, we studied public videos and repositories of other DataQuest 2025–2026 teams; what we adopted, with credit: README → Acknowledgements.

METHODS & ETHICS

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AI ASSISTANCE & REPRODUCIBILITY

- AI assistance: Claude Code (Anthropic) AI agents, directed by the team, wrote most of the R code, the memos, the figures and the slide and script text; team members voice the video (no synthetic voice). Every number in the text and narration was recomputed from the raw UN files by separate code (also AI-written) and checked by hand by the four team members; numbers inside figures were recomputed by that code. The four of us share equal responsibility. Log: docs/registro_ia.md.
- Code, notebook, figures and hypotheses (fixed 2026-10-23 before the final run, after exploring the same data — not confirmatory; sha256 83f98f6e1a3d) are in our public repository **https://github.com/DarkNightSoldier/ASA_DataQuest_2026** (code MIT; texts CC BY 4.0).